

$$\underline{\underline{f(x)}} = \underline{\underline{a_0}} + \underline{\underline{a_1(x-k)}} + \underline{\underline{a_2(x-k)^2}} + \dots + a_n(x-k)^n + \dots$$

put $x=k$.

$$f(k) = a_0,$$

$$f'(x) = a_1 + 2a_2(x-k) + \dots$$

put $x=k$

$$f'(k) = a_1$$

$$f''(x) = 2a_2 + 6a_3(x-k) + \dots$$

$$f''(k) = 2a_2$$

$$a_2 = \frac{1}{2}f''(k)$$

$$f^{(n)}(x) =$$

$$\rightarrow a_n(x-k)^n$$

$$1 \rightarrow na_n(x-k)^{n-1}$$

$$2 \rightarrow n(n-1)a_n(x-k)^{n-2}$$

$$\textcircled{3} \rightarrow \underline{\underline{n(n-1)(n-2)a_n(x-k)^{n-3}}}$$

$$n \rightarrow \underline{\underline{(n! a_n)}}$$

$$f^{(n)}(x) = n!a_n + \underline{\underline{\binom{n}{1}(x-k) + \binom{n}{2}(x-k)^2 + \dots}}$$

$$f^{(n)}(k) = n!a_n$$

$$\Rightarrow a_n = \frac{f^{(n)}(k)}{n!}$$

① e^x .

(i) expand at $x=0$.

$$\underline{e^x} = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$$

$$= f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots + \frac{f^{(n)}(0)}{n!} x^n + \dots$$

$$= 1 + \frac{1}{1!} x + \frac{1}{2!} x^2 + \dots + \frac{1}{n!} x^n + \dots$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

(ii) expand at $x=2$.

$$a_0 + a_1(x-2) + a_2(x-2)^2 + \dots + a_n(x-2)^n + \dots$$

$$a_n = \frac{f^{(n)}(2)}{n!}$$

$$= f(2) + \frac{f'(2)}{1!} (x-2) + \frac{f''(2)}{2!} (x-2)^2 + \dots$$

$$= e^2 + \frac{e^2}{1!} (x-2) + \frac{e^2}{2!} (x-2)^2 + \dots + \frac{e^2}{n!} (x-2)^n + \dots$$

$$(2) \ln(1+x), \quad f'(x) = \frac{1}{1+x}, \quad f''(x) = -\frac{1}{(1+x)^2}, \quad f'''(x) = \frac{2}{(1+x)^3}, \quad f^{(iv)}(x) = -\frac{6}{(1+x)^4}$$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(iv)}(0)}{4!}x^4 + \dots$$

$$= 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \quad \frac{(n-1)!}{n!} = \frac{1}{n}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\lim_{x \rightarrow 0} \ln(1+x) \simeq x, \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

$$\textcircled{3} \sin x, \quad f(x) = \underline{\underline{\cos x}}, \quad f'(x) = -\underline{\underline{\sin x}}, \quad f''(x) = -\underline{\underline{\cos x}}, \quad f^{(4)}(x) = \sin x$$

$$\sin x = f(0) + \frac{f'(0)}{1!}x + \frac{1}{2!}f''(0) \cdot x^2 + \frac{1}{3!}f^{(3)}(0) \cdot x^3 + \dots$$

$$= 0 + \underset{\uparrow}{x^1} + 0 - \underset{\uparrow}{\frac{x^3}{3!}} + 0 + \frac{x^5}{5!} - \dots$$

$$= \underline{\underline{x}} - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

① $V = e^{-yT}$, $(y) \rightarrow y + \Delta y$ d, Δ^f

$\Delta \rightarrow 0 \rightarrow d$
 difference differential

$$\Delta f = f'(x_0) \cdot (\Delta x) + \frac{1}{2} f''(x_0) (\Delta x)^2$$

$$\Delta V = \left(\frac{dV}{dy} \right) (\Delta y) + \frac{1}{2} \left(\frac{d^2V}{dy^2} \right) (\Delta y)^2$$

$$\frac{dV}{dy} = e^{-yT} \cdot (-T), \quad \frac{d^2V}{dy^2} = (e^{-yT}) \cdot T^2$$

$$\Delta V = -e^{-yT} \cdot T \cdot (\Delta y) + \frac{1}{2} e^{-yT} \cdot T^2 \cdot (\Delta y)^2$$

$$= -V \cdot T \cdot (\Delta y) + \frac{1}{2} V \cdot T^2 \cdot (\Delta y)^2$$

$$\Rightarrow \frac{\Delta V}{V} = \underbrace{-T \cdot (\Delta y)}_{\text{modified duration}} + \underbrace{\frac{1}{2} T^2 \cdot (\Delta y)^2}_{\text{convexity}}$$

(%) (convexity)

$$\Delta y = +ve$$

$$(-ve)$$

$$(2) V = \frac{F_c}{(1+y)} + \frac{F_c}{(1+y)^2} + \dots + \frac{F_c}{(1+y)^T} + \frac{F}{(1+y)^T}$$

$$\begin{aligned} \frac{dV}{dy} &= -\frac{F_c}{(1+y)^2} - \frac{2F_c}{(1+y)^3} - \dots - \frac{TF_c}{(1+y)^{T+1}} - \frac{TF}{(1+y)^{T+1}} \\ &= -\left(\frac{1}{1+y}\right) \left(1 \times \frac{F_c}{(1+y)} + 2 \times \frac{F_c}{(1+y)^2} + \dots + T \times \frac{F_c}{(1+y)^T} + \frac{TF}{(1+y)^T}\right) \end{aligned}$$

$$= -\frac{V}{1+y} D$$

$$\Delta V = \left(\frac{dV}{dy}\right) \cdot \Delta y = -\left[\frac{V}{1+y} D\right] (\Delta y) = -V \cdot \underset{\substack{\text{Macaulay} \\ \text{duration}}}{D} \cdot \underset{\substack{m \text{ Duration} \\ \text{(modified)}}}{D^*} \cdot \Delta y$$

$$\left(\frac{\Delta V}{V}\right) = -\underset{\uparrow}{D^*} \cdot \Delta y$$

$$D^* = \frac{D}{1 + y/m}$$

$$\langle D^* = D \rangle$$

m - compounding frequency

$m \rightarrow \infty$ (Continuous compounding)